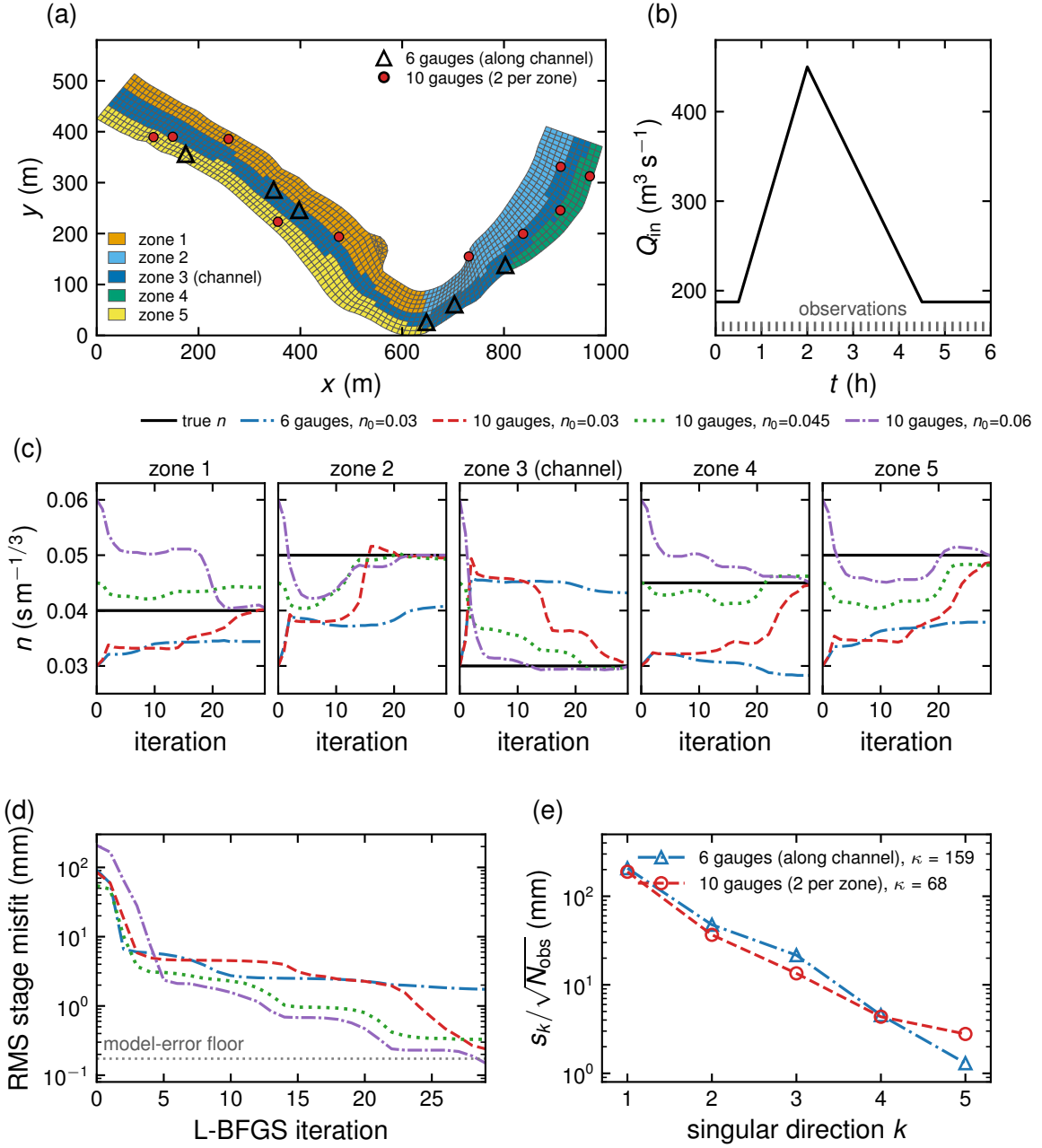

(a)
6 gauges (along channel)
10 gauges (2 per zone)
zone 1
zone 2
zone 3 (channel)
zone 4
zone 5
y (m)
x (m)
(b)
Q_{in} (m^3 s^{-1})
observations
t (h)
true n
6 gauges, n_0=0.03
10 gauges, n_0=0.03
10 gauges, n_0=0.045
10 gauges, n_0=0.06
(c)
zone 1
zone 2
zone 3 (channel)
zone 4
zone 5
n (s m$^{-1/3}$)
iteration
(d)
RMS stage misfit (mm)
model-error floor
L-BFGS iteration
(e)
6 gauges (along channel), κ = 159
10 gauges (2 per zone), κ = 68
$s_k/\sqrt{N_{obs}}$ (mm)
singular direction k